Today's Agenda

Activation Functions

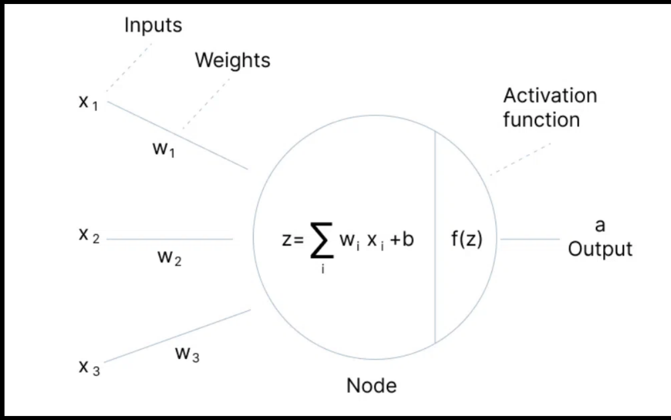
Optimizers

Next -> Loss Func

Last Class

Forward & Back Prop

Activation Functions



* Input -> Choices { RELU }
* Hidden -> Choices { RELU }
* Output -> Sigmoid
  Softmax

Why Activation? (without)

Linear Transformation w.r.t Weight & Bias

Composition of 2 Linear Functions

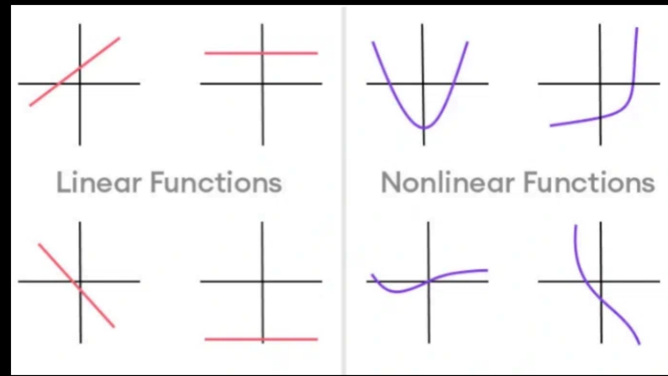
⇓

Linear

* Non-Linearity *

1) Positive Gradient

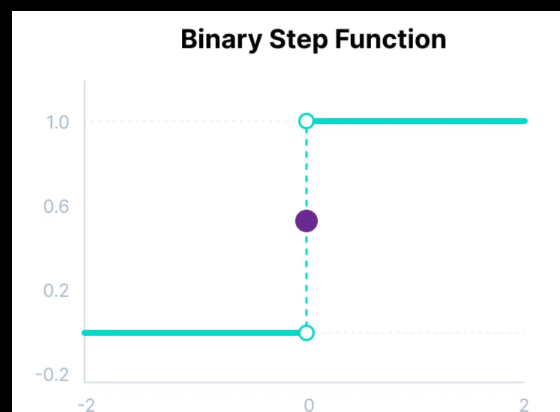
2) Negative Gradient



Influence

Step Function

$$f(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 & \text{for } x \geq 0 \end{cases}$$



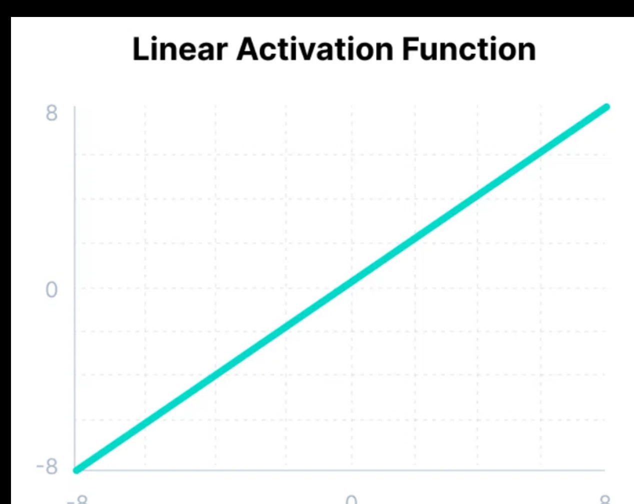
Gradient of a step function is zero (0).

Linear Activation

No Activation

Identity Function

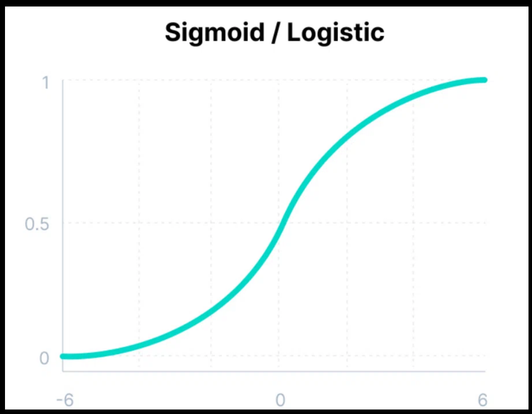
$$f(x) = x$$

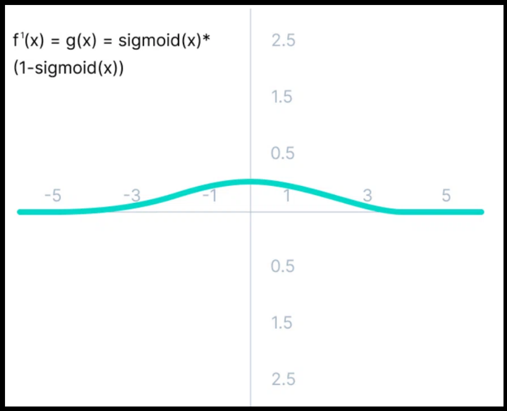


Non-Linear Activations

1) Logistic Activation (Sigmoid)

0 - 1





-3 to 3

gradient

↓

0

$$\text{Sigmoid } f(x) = \frac{1}{1+e^{-x}}$$

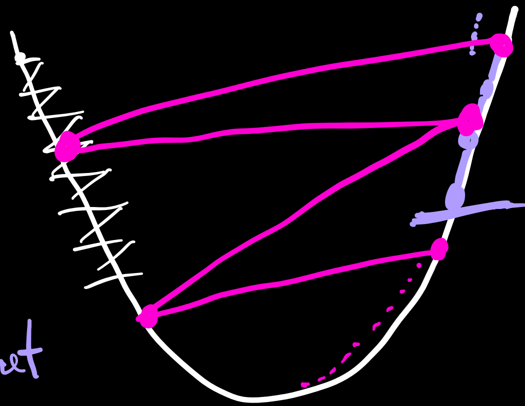
$$f'(x) = \text{sigmoid}(x) * (1 - \text{sigmoid}(x))$$

Vanishing Gradient

$W_{new} \approx W_{old}$ (No learning) ✗

$$\left\{ W_{new} = W_{old} - \eta \frac{\partial L}{\partial w_{old}} \right\}$$

$$W_{old} - \underline{.0000000000001} \times .0001$$



Vanishing Gradients

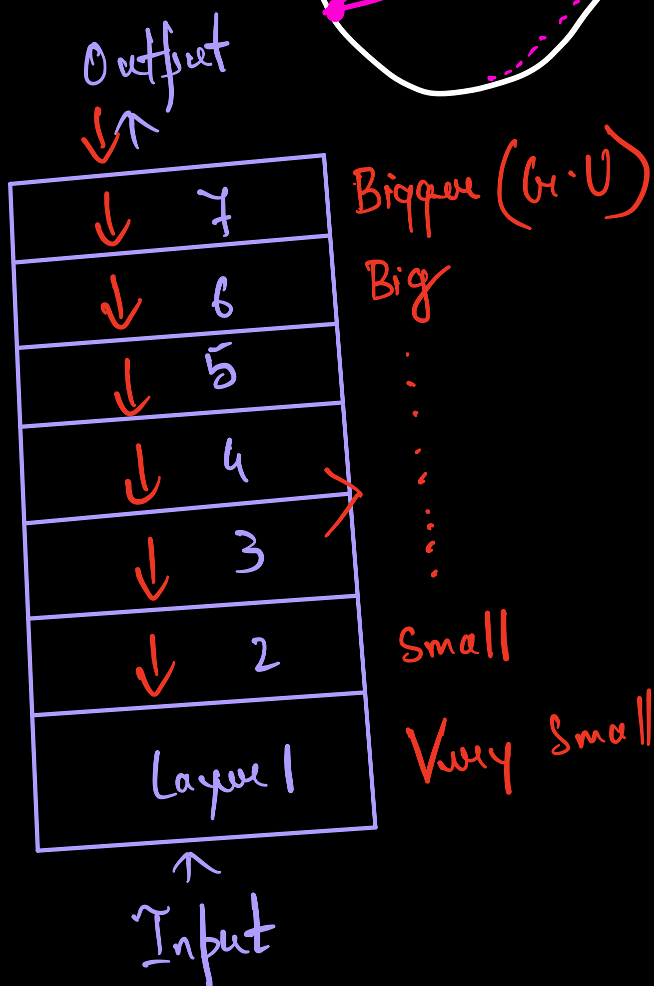
Networks become deeper.

50 layer

100 layer

150 layers

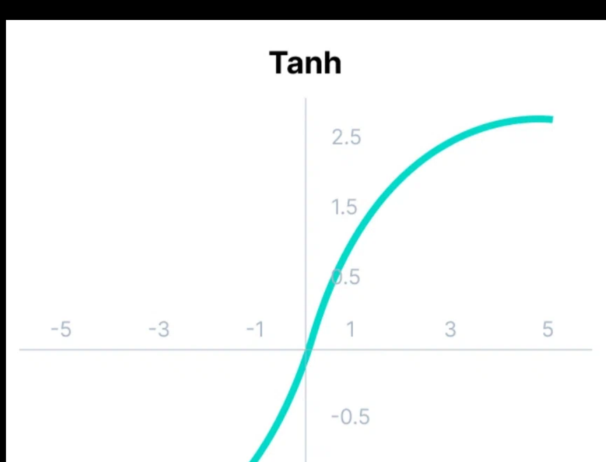
200 Layers



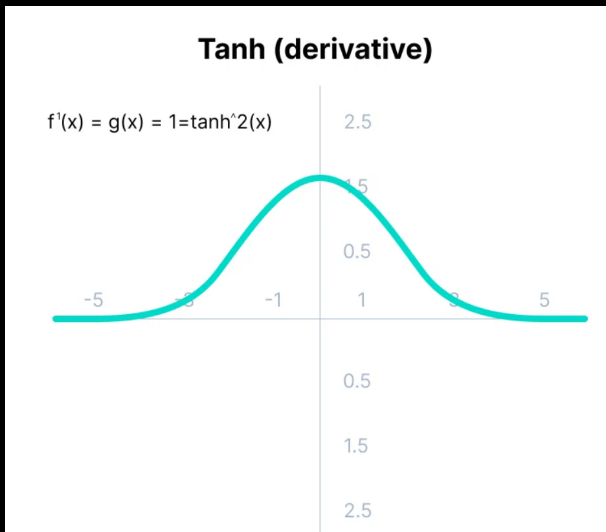
$1 = 1.00000007$

Residual Connection

Tanh Activation (Range



$$f(x) = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}$$

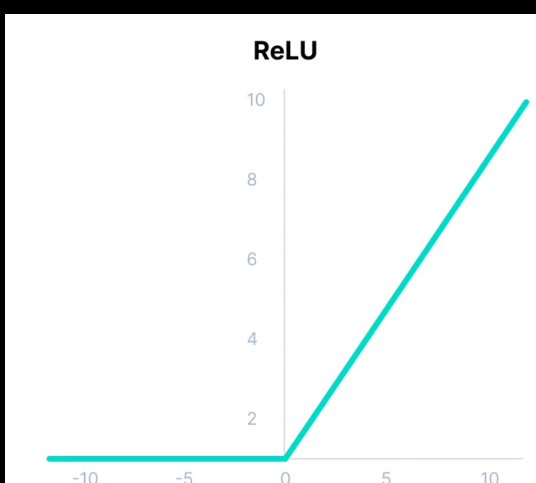


$$f'(x) = 1 - \tanh^2(x)$$

1) Highly +ve
2) Neutral (0 centric)
3) Highly -ve

V.G.D

RELU (Rectified Linear Unit)

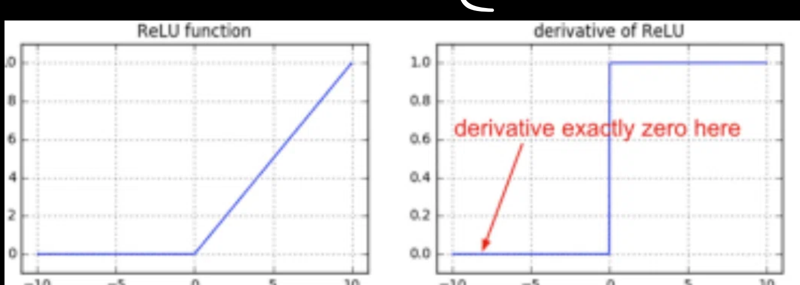


$$f(x) = \max(0, x)$$

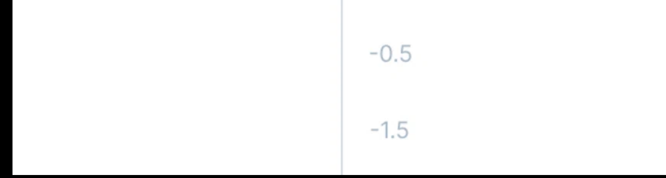
Computationally Efficient

Easy

{Hidden layers}



Dead Neurons

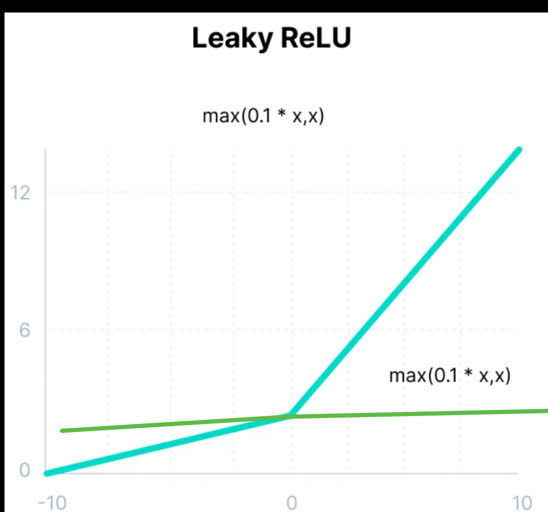


1) Model fitting

H1 - 100
Shuffle of
neurons.

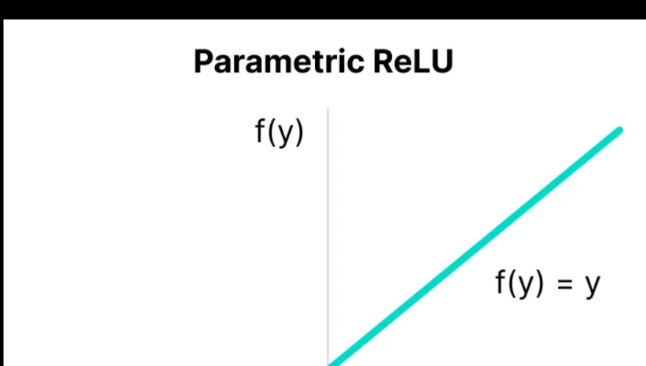
Dying Relu Problem

Leaky Relu

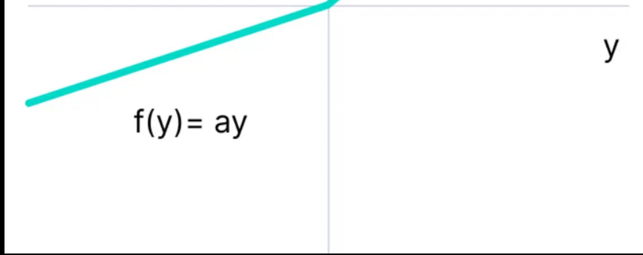


$$f(x) = max(0.1x, x)$$

Parametric Relu

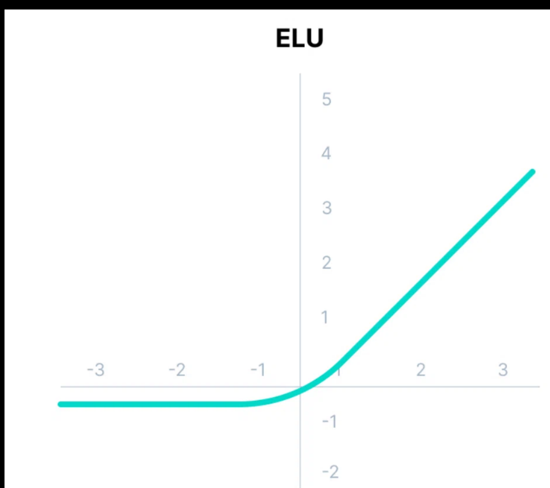


$$f(x) = max(ax, x)$$



$a =$ parameter

ELU (Exponential Linear Units)



$$f(x) = \begin{cases} x & \text{for } x \geq 0 \\ \alpha(e^{x} - 1) & \text{for } x < 0 \end{cases}$$

No learning in α

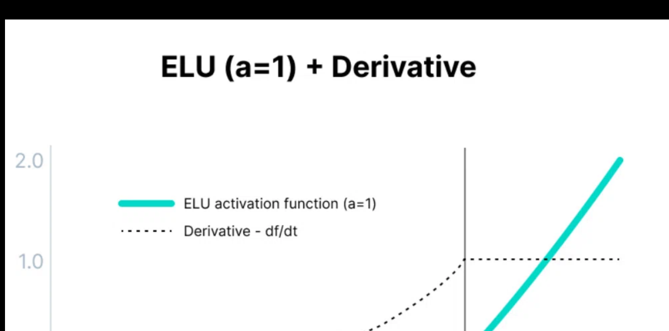
Exploding Gradient Problem

large $\leftarrow W_{new} \ggg W_{old} \rightarrow$ small

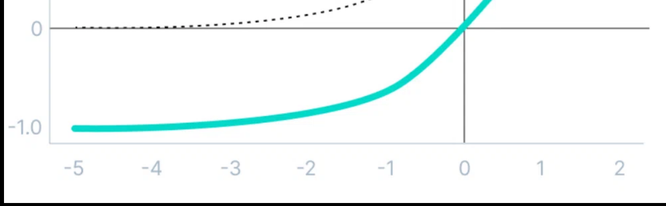
big

$$W_{new} = W_{old} - \left\{ \eta \frac{\partial L}{\partial w_0} \right\}$$

Very high



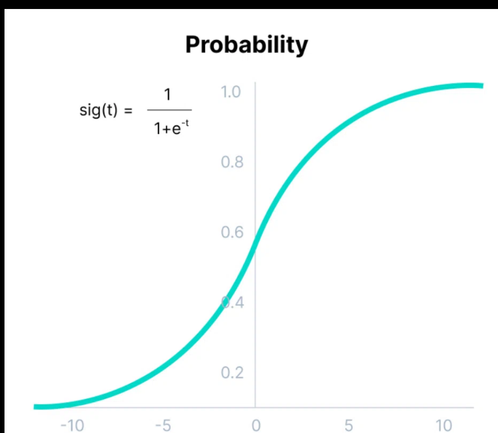
$$f'(x) = \begin{cases} 1 & x \geq 0 \\ f(x) + \alpha & x < 0 \end{cases}$$



Softmax (Activation)

Range of the prob - (0 - 1)

Sum of all the probab = 1



0.8 , 0.7 , 0.6 , 0.5 , 0.2

(class = z_1, z_2, z_3
3 class

$$\text{Softmax}(z_1) = \frac{e^{z_1}}{e^{z_1} + e^{z_2} + e^{z_3}}$$

$$(z_2) = \frac{e^{z_2}}{e^{z_1} + e^{z_2} + e^{z_3}}$$

$$(z_3) = \frac{e^{z_3}}{e^{z_1} + e^{z_2} + e^{z_3}}$$

Multi-class classification

$$z_1 + z_2 + z_3 = 1$$

Relative Prob

$$\text{Softmax}(z_i) = \frac{\exp(z_i)}{\quad}$$

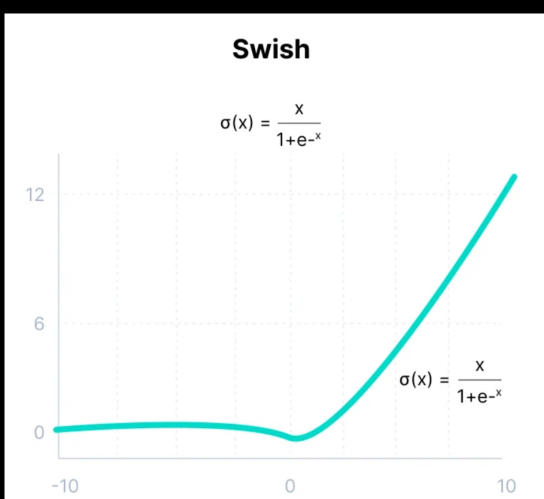
$\sum_j \exp(z_j)$

Output layer

Probability = [1.8, 0.9, 0.68]

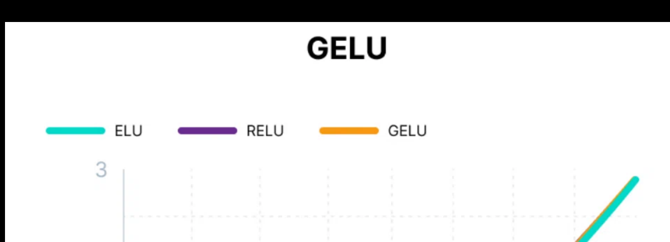
Softmax = [.577, 0.235, 0.188]

Swish Activation



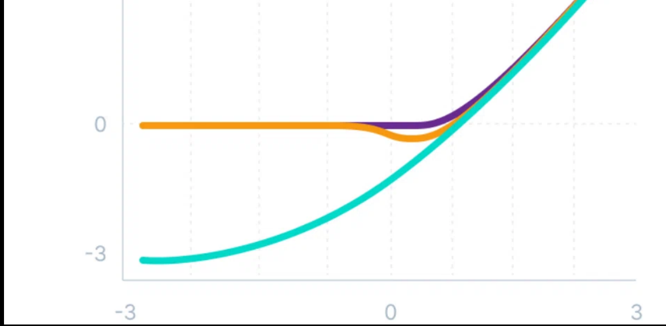
$f(x) = x * \text{sigmoid}(x)$

GELU → Gaussian Error Linear Unit



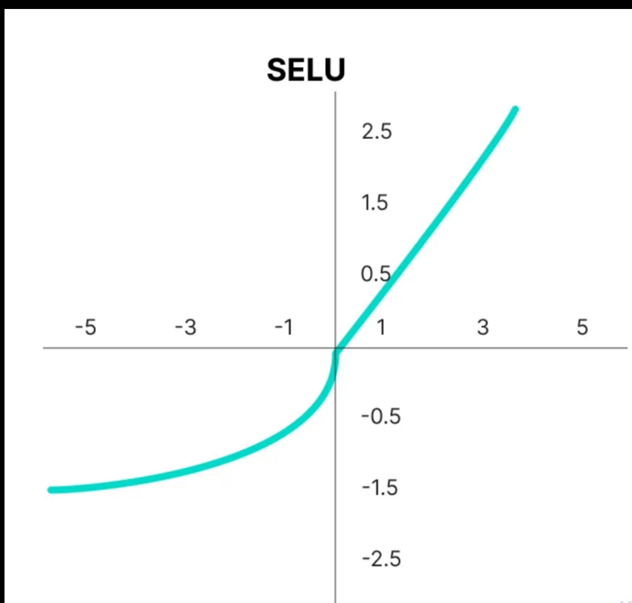
GELU

$$f(x) = xP(X \le x) = x\Phi(x)$$



$$f(x) = xP(X \le x) = x\Phi(x)$$
$$= 0.5x\left(1 + tanh\left[\sqrt{2/\pi}\left(x + 0.044715x^3\right)\right]\right)$$

Scaled ELU



SELU

$$f(\alpha, x) = \lambda \begin{cases} \alpha(e^x - 1) & for\ x < 0 \\ x & for\ x \geqslant 0 \end{cases}$$

Internal Normalization

Mean & Variance

of the previous layer.

Internal Norm

$\Downarrow$

Convergence speed higher